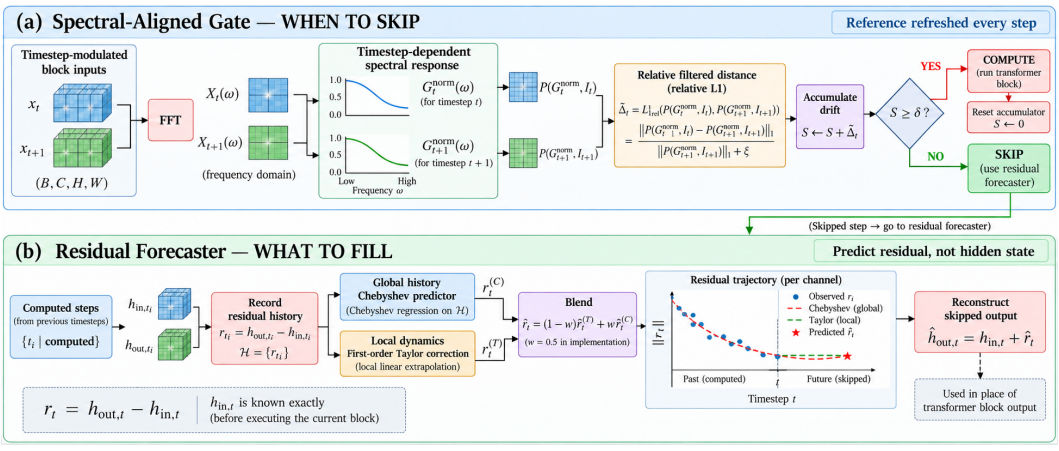

(a) Spectral-Aligned Gate — WHEN TO SKIP
Reference refreshed every step
Timestep-modulated block inputs
x_t
x_{t+1}
(B, C, H, W)
FFT
$X_t(\omega)$
$X_{t+1}(\omega)$
(frequency domain)
Timestep-dependent spectral response
1.0
0.5
0.0
$G_t^{\text{norm}}(\omega)$
(for timestep t)
$G_{t+1}^{\text{norm}}(\omega)$
(for timestep $t+1$)
Low
High
Frequency ω
$P(G_t^{\text{norm}}, I_t)$
$P(G_{t+1}^{\text{norm}}, I_{t+1})$
Relative filtered distance (relative L1)
$\tilde{\Delta}_t = L1_{\text{rel}}(P(G_t^{\text{norm}}, I_t), P(G_{t+1}^{\text{norm}}, I_{t+1}))$
$= \frac{\|P(G_t^{\text{norm}}, I_t) - P(G_{t+1}^{\text{norm}}, I_{t+1})\|_1}{\|P(G_{t+1}^{\text{norm}}, I_{t+1})\|_1 + \xi}$
Accumulate drift
$S \leftarrow S + \tilde{\Delta}_t$
$S \geq \delta$?
YES
COMPUTE (run transformer block)
Reset accumulator $S \leftarrow 0$
NO
SKIP (use residual forecaster)
(Skipped step → go to residual forecaster)
(b) Residual Forecaster — WHAT TO FILL
Predict residual, not hidden state
Computed steps (from previous timesteps)
$\{t_i \mid \text{computed}\}$
h_{in,t_i}
h_{out,t_i}
Record residual history
$r_{t_i} = h_{\text{out},t_i} - h_{\text{in},t_i}$
$\mathcal{H} = \{r_{t_i}\}$
Global history Chebyshev predictor
(Chebyshev regression on $\mathcal{H}$)
$r_t^{(C)}$
Local dynamics First-order Taylor correction
(local linear extrapolation)
$r_t^{(T)}$
Blend
$\hat{r}_t = (1-w)\hat{r}_t^{(T)} + w\hat{r}_t^{(C)}$
($w = 0.5$ in implementation)
Residual trajectory (per channel)
$\|r_t\|$
Observed r_t
Chebyshev (global)
Taylor (local)
Predicted $\hat{r}_t$
Past (computed)
t
Future (skipped)
Timestep t
Reconstruct skipped output
$\hat{h}_{\text{out},t} = h_{\text{in},t} + \hat{r}_t$
Used in place of transformer block output
$r_t = h_{\text{out},t} - h_{\text{in},t}$
$h_{\text{in},t}$ is known exactly (before executing the current block)